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Tinnitus Pitch: A Comparison of Three Measurement Methods*

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Abstract

The most prominent pitch of tinnitus was measured in 10 subjects with sensorineural tinnitus. The pitch was determined with three different psychophysical procedures in the ear *ipsilateral* to the tinnitus; an *Adaptive Method* (Bracketing), a *Method of Limits* (ascending and descending), and the *Method of Adjustment*. Each procedure involved equating the pitch of a pure tone to the most prominent tinnitus pitch, and was repeated seven times on each subject. Although there was no statistically significant difference for the means and standard deviations among the different methods for the group data, there were some large differences in a few individuals. Many of the subjects produced pitch matches that covered a range of 1 octave, whereas others showed better consistency. The Method of Limits took longer to perform and resulted in more octave confusions than the other two methods. The Adaptive Method was also repeated five times for each subject in the ear *contralateral* to the tinnitus. Two subjects produced a tinnitus pitch match that was over ½ octave lower in the contralateral ear. We recommend that tinnitus pitch be measured in the ipsilateral ear with either the Method of Adjustment or the Adaptive Method. Because some patients are unreliable in their pitch matching we suggest repeating the match seven to nine times.

Introduction

Although the handicap created by tinnitus is great (Tyler and Baker, 1983), very little methodological work has gone into its measurement. Tinnitus pitch or loudness could: (1) have diagnostic significance (e.g. Nodar and Graham, 1965; Douek and Reid, 1968), (2) provide a quantitative measure to monitor tinnitus deterioration or improvement during therapy (Donaldson, 1978), and (3) provide part of a classification scheme necessary for its study.

From subjective patient reports it appears that tinnitus can be composed of many different types of sounds. For example, a patient might describe his tinnitus as containing a low-pitch buzz and a high-pitch screeching noise. Our approach, here and in clinical evaluations, is to ask patients to concentrate on the predominant or main pitch of their tinnitus. Virtually all the patients that we have seen are content performing the pitch match to the predominant pitch of their tinnitus. We therefore define 'tinnitus pitch' operationally as the frequency of an acoustic tone that has been matched to the predominant pitch of the tinnitus. The acoustic tone does not necessarily sound like the tinnitus. Nevertheless it does provide a quantifiable metric that we presume relates to the internal spectral or temporal characteristics of the tinnitus and to the pitch-like quality of the tinnitus.

Previous reports of tinnitus pitch matching have generally used some form of 'bracketing' technique (Mortimer *et al.*, 1940; Fowler, 1940; Graham and Newby, 1962; Douek and Reid, 1968; Shailer *et al.*, 1981). Unfortunately, this procedure is rarely defined with sufficient precision to determine its reliability or to allow others to replicate it precisely. Some researchers have tested the ear contralateral to the tinnitus (e.g. Fowler, 1940; Reed. 1960), some ipsilateral to the tinnitus (e.g. Roeser and Price, 1980), and some in free field (Mortimer *et al.*, 1940). In a few papers the test ear is not even reported. Many patients with tinnitus will have a hearing loss and possibly binaural diplacusis, which could confound the interpretation of the tinnitus pitch. Therefore it is not clear whether ipsilateral pitch-matching results are compatible with contralateral results.

In the present study we have compared three different psychophysical methods for measuring tinnitus pitch, and have performed one of these methods in both the ipsilateral and contralateral ear.

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Method

Subjects

All subjects had sensorineural tinnitus (i.e. they had a sensorineural hearing loss and no indication of a middle-ear pathology or middle-ear tinnitus source). There were three female and seven male subjects, whose average age was 56.1 years (s.d. = 10.6). All subjects stated that the pitch of their tinnitus was steady when present. Although tinnitus can frequently be composed of many different types of sounds, we instructed the subjects to concentrate on the *predominant* pitch of their tinnitus. We did not pre-select subjects on the basis of their description of their tinnitus. The tinnitus ear was defined as the ear, or the side of the head, judged to contain the tinnitus. Seven subjects localised their tinnitus distinctly to one ear or one side of the head. Subjects 1 and 4 had a bilateral tinnitus that was louder on the left side. Subject 9 had a diffuse tinnitus that appeared to be located throughout the head, and in this case the right ear was arbitrarily chosen as the ipsilateral ear. Subject 4 judged the tinnitus to be of equal loudness in the two ears, but reported that the tinnitus in his left ear was 'more noticeable'.

Thresholds for bone-conduction and air-conduction signals suggested the absence of a conductive hearing loss in all subjects. Three subjects (1, 2 and 5) had pure-tone thresholds between 0 and 20 dB HL. Four subjects (4, 6, 7 and 10) had low-frequency (0.5 and 1.0 kHz) thresholds < 10 dB HL together with elevated high-frequency thresholds. The remaining three subjects (3, 8 and 9) typically had low-frequency pure-tone thresholds > 20 dB HL together with a more pronounced high-frequency threshold loss.

Apparatus

The pure tone was generated by a Wavetec VCG/Noise Generator 132, and led to a Klark-Technik DN 27 Graphic Equalizer. This contains a bank of 27 $\frac{1}{3}$-octave filters whose output level can be adjusted over a 24 dB range. The output levels of these filters were set individually for each subject so that the pure tone remained audible and comfortable throughout the frequency range. The output was led to a manual attenuator (Marconi TF 21 62), and a second attenuator with 1 dB step sizes that covered a 30 dB range. This was followed by a signal gate (Coulbourn S4-04), which was controlled by a timer (Coulbourn S3-21). The rise–fall times of the signal were 40 ms, and the signal was repetitively pulsed with a 1 s on-and-off time. Harmonic levels of the pure tones were at least 55 dB below the primary level throughout the frequency and amplitude range tested. The output of the gate was led to a Telephonics TDH-49 earphone encased in an MX-41/AR cushion. Testing was performed in a single-walled I.A.C. booth enclosed within a concrete sound-isolating enclosure. In the Adaptive Method and the Method of Limits, the tester set the signal frequency within 1% of the nominal value.

Procedure

First, the concept of low and high pitch was demonstrated by allowing the subject to listen to low-frequency and high-frequency tones. For each method, the subject practised for about 5–10 min until the tester was assured that the task was understood. Con slidesrs

In the *Method of Adjustment*, the subject was instructed: 'Adjust the pitch of the pulsing tone using this dial. Make wide sweeps with the dial at first then gradually centre in on the pitch which is nearest to your tinnitus'. A cardboard overlay was placed over the dial to prevent the subject from obtaining visual clues from the dial markings. The subject first manipulated the main frequency dial of an oscillator that presented pulsed tones, and was then encouraged to make further adjustments with the fine-control dial. Metodo propuesto

In the *Method of Limits* the subject was instructed: 'State whether your tinnitus is "higher" or "lower" in pitch than the tone that I present'. The tester then presented the subject with a sequence of pulsed tones and allowed the subject to make a 'higher' or 'lower' decision. On an ascending run the first frequency tested was 125 Hz and this frequency was then increased in $\frac{1}{8}$-octave steps until the subject's response changed from 'higher' to 'lower'. For subject 2, where the initial response was 'lower', the starting frequency was set at 40 Hz. On a descending run the initial frequency was 8000 Hz and the tone frequency decreased in $\frac{1}{8}$-octave steps until the response changed from 'lower' to 'higher'. For two subjects where the initial response was 'higher', the starting frequency was 14,000 Hz. An ascending run was always accompanied by a descending run, although the order of presentation was randomised. In the ascending run the tinnitus pitch was taken as the last 'higher' response, and in the descending run it was taken as the last 'lower' response. The overall tinnitus pitch was then taken as the average of these two values. Metodo mas comodo para el sujeto

In the *Adaptive Method* the subject was instructed as in the Method of Limits. Again the tester presented the subject with a sequence of pulsed tones and allowed the subject to make a response of 'higher' or 'lower' after each presentation. In the first stage, the tinnitus pitch was located to within a 1-octave band for test frequencies of 63, 125, 250, 500, 1000, 2000, 4000, and 8000 Hz, by successively eliminating extreme frequency values from the range. The initial tone frequency was either 125 Hz or 8000 Hz (chosen in random order). If the subject responded that his tinnitus was 'lower' than the tone, then the frequency was decreased to the lowest test frequency that had not already been presented. If the subject responded that his tinnitus was 'higher' than the tone, then the frequency was increased to the highest test frequency that had not already been presented. For subject 7 the response to 8000 Hz was 'higher', and therefore an additional test frequency of 14,000 Hz was introduced. In the second stage tones were presented whose frequencies were within the 1-octave range determined from Stage 1. The tinnitus pitch was located within a $\frac{1}{8}$-octave range (ISO 266-1975 E) and the tinnitus pitch was arbitrarily taken as the higher value of the final $\frac{1}{8}$-octave range tested (see Tyler, 1982; Fig. 7).

When the tinnitus pitch had been determined a tone was presented to the test ear one octave below and one octave above the pitch-match frequency to test for octave confusions. The subject was asked to indicate if the pitch of this tone was nearer to the tinnitus pitch.

Each method was performed seven times with the matching tone in the ipsilateral ear. For the Method of Limits, this includes seven ascending and seven descending trials. The Adaptive Method was also performed five times with the matching tone in the contralateral ear. The methods were performed in random order.

Subjects were also asked to estimate their tinnitus pitch on a scale of 1 to 6 (1 being very low and 6 being very high).

Results

When averaging pitch-matching data, one has the option of using either a linear or logarithmic scale. We have analysed our data in both ways, and for the sake of brevity and ease of interpretation we present results using the linear scale. However, analyses of the data using a logarithmic scale resulted in identical conclusions. Our data were analysed with and without changes due to octave confusions. Since the results of our analysis based on both sets of data were identical, we present only the data where changes due to octave confusions were used.

Comparison across methods

Table I displays the individual results obtained from the different methods, with the average results shown at the bottom. Overall there is no statistically significant difference (ANOVA) among methods for the pitch matches or their standard deviations (s.d.). However, there are large differences among the methods for a few individuals. Subject 9, for example, displays a pitch match with the Method of Limits that is more than one octave lower than the pitch match obtained with the Method of Adjustment. Subject 8 displays a pitch match with the Adaptive Method that is nearly one octave lower than the pitch-match estimate with the Method of Limits.

There were also notable individual differences for the s.d.s. Subject 2 had an s.d. of 13 with the Method of Adjustment and an s.d. of 4 with the Method of Limits. Subject 4 had an s.d. of 910 with the Method of Limits and an s.d. of 3958 with the Adaptive Method. The range of pitch matches covered more than a one-octave span in six subjects on the Method of Adjustment, four subjects on the Method of Limits and four subjects on the Adaptive Method (Ipsilateral). On the other hand, other subjects were much more precise. For example, with the Method of Adjustment subject 5 produced seven matches between 7894 and 9037 Hz. With the Method of Limits subject 7 produced seven matches between 9000 and 10,600 Hz, and with the Adaptive Method subject 2 produced seven matches within 50 and 56 Hz.

For the Method of Limits, the results of the ascending and descending approach were averaged. The average is a preferable measure as it reduces errors of expectation and habituation (e.g. Corso, 1967). In eight of the ten subjects the ascending approach produced a lower pitch match than did the descending approach, and the difference between the ascending and descending results was statistically significant ($P < 0.05$). In three subjects (3, 4 and 10), the pitch match with the ascending approach was over two octaves lower in frequency than the pitch match with the descending approach.

The results obtained with the Adaptive Method performed in the contralateral ear are shown on the right of Table I. Although there was no statistically significant difference between the pitch-match frequency obtained for the ipsilateral and contralateral conditions, there are some notable individual differences. Subjects 3 and 6 produced a pitch match that was over $\frac{1}{2}$ octave lower in the contralateral ear. There was no significant difference between the ipsilateral and contralateral standard deviations.

Octave confusions

Twenty-six out of a possible 330 pitch matches (7.8%) resulted in octave confusions (Table II). Very few octave confusions occurred for the Adaptive Method, and none was observed for the

Table I. Individual pitch-matching results are shown for the mean, standard deviation [s.d.; using (n − 1)] of the pitch matches, and range. Seven replications were performed for the ipsilateral procedures and five replications were performed for the contralateral procedure

	Adjustment (ipsilateral)			Limits (ipsilateral)		
Subject number	Pitch-match frequency (Hz)	s.d.	Range	Pitch-match frequency (Hz)	s.d.	Range
1	7750	2750	4071–10,451	4073	1596	2695–7375
2	57	13	38–75	49	4	40–54
3	387	196	256–826	1079	351	672–1512
4	7386	2233	4821–10,376	4483	910	3115–5560
5	8597	406	7894–9037	8057	486	7300–8500
6	411	101	327–591	383	33	335–450
7	10,302	1373	8569–11,724	9814	520	9000–10,600
8	917	342	690–1565	1602	548	960–2750
9	324	121	214–528	139	19	126–182
10	3357	1717	1243–5217	1895	266	1460–2055
Average	3948	925		3157	473	

	Adaptive (ipsilateral)			Adaptive (contralateral)		
Subject number	Pitch-match frequency (Hz)	s.d.	Range	Pitch-match frequency (Hz)	s.d.	Range
1	4928	1600	2800–7100	3250	760	2000–4000
2	52	3	50–56	55	5	50–63
3	667	326	400–1000	384	51	315–450
4	5145	3958	900–11,200	7600	5672	1000–12,500
5	7728	1620	5600–10,000	8600	894	8000–10,000
6	457	44	400–500	275	26	250–315
7	12,742	981	11,200–14,000	10,480	657	10,000–11,200
8	728	273	400–1120	1040	305	560–1400
9	239	46	160–280	251	28	224–280
10	1114	665	500–2240	1326	526	900–2240
Average	3380	951		3326	892	

50 pitch matches performed in the contralateral ear. Most of the octave confusions occurred with the Method of Limits. Subject 1 made 7 octave confusions out of 7 pitch matches with the descending Method of Limits. Considering all the methods, 21 changes were made to a lower frequency, and 5 changes were made to a higher frequency. In about a quarter of the pitch matches we were unable to present a tone one octave above the pitch-match frequency at a level audible to the subject.

Scaled tinnitus pitch

We found no relationship between scaled pitch and pitch-matching measurements. The Pearson correlation between the subjects' scaled tinnitus pitch (between 1 and 6) and the logarithm of their tinnitus pitch matches was not significant for all conditions that we tested. Subject 9, for example, estimated his tinnitus pitch as a 6, but his pitch-match frequency was 37 Hz. We suspect this poor relationship is due to different individuals having different concepts of what constitutes a low and a high pitch, and to the probability that the 1 to 6 category scale does not have the properties of a ratio scale (see Stevens, 1975).

Discussion

The large individual differences that were observed, the inter-ear difference seen in some listeners, and the presence of octave confusions provide some insights into tinnitus pitch perception and permit us to offer some guidelines about its measurement.

Differences between contralateral and ipsilateral measurements

There are several possible reasons why ipsilateral/contralateral differences might exist.

Table II. Octave confusions observed in all methods. The total number of pitch matches performed with each method is shown in the first column and the number of subjects (out of ten) who committed octave confusions is shown in the second. The number of pitch matches that were altered to a lower pitch and to a higher pitch is shown in the next two columns. The total octave confusions for each method are shown in the last column

	Number of pitch matches	Subjects changing pitch-match	Changes to lower pitch	Changes to higher pitch	Total octave confusions
Adaptive					
Ipsilateral	70	2	2	0	2
Contralateral	50	0	0	0	0
Limits (ipsilateral)					
Ascending	70	4	0	5	5
Descending	70	3	12	0	12
Adjustment (ipsilateral)	70	5	7	0	7
Total	330	8	21	5	26

First, some of the differences might result from level differences of the matching tones. However, in normal listeners the effect of level on pitch of pure tones is usually less than 5% (e.g. Stevens, 1935; Small and Campbell, 1961; Terhardt, 1974a), and we suspect this has a limited effect on our results. Second, (a) the tinnitus may alter the pitch of the pure tone, or (b) the tone may alter the pitch of the tinnitus or (c) both. For example the presence of narrow-band noise can alter the pitch of a pure tone (Webster and Schubert, 1954). Several authors (Minton, 1923; Jones and Knudson, 1928; Josephson, 1931; Fowler, 1938; Douek and Reid, 1968) have suggested that tinnitus can interfere with the detection of the pure tone, although a dip in the audiogram in the frequency region of the pitch match might be a direct result of the pathology. Third, a combination tone (e.g. Greenwood, 1972) might arise between the pure tone and the tinnitus and this might result in an additional pitch sensation. Fourth, monaural and binaural beats (at least at frequencies below 1500 Hz; Licklider *et al.*, 1950) between a 'tonal' tinnitus and the pure tone might also influence ipsilateral/contralateral differences. Fowler (1940, 1941, 1942) has suggested that beats do not occur between a pure tone and tinnitus, although a few subjects do report beating (Wegel, 1931; Flottorp, 1953; Stanaway *et al.*, 1970; Lackner, 1967). Josephson (1931), Lackner (1967), Shailer *et al.* (1981) and Tyler and Conrad-Armes (1983) have reported individuals with tinnitus where a lateralisation-like effect was noted (where the tinnitus appears to move inside the head), but the frequency specificity of the effect is unknown.

Finally, several authors have reported that a pure tone presented to the two different ears of an individual can produce two different pitch sensations. This binaural diplacusis has been noted in individuals with abnormal (Shambaugh, 1940; Gaeth and Norris, 1965; van den Brink, 1970) or normal pure-tone thresholds (Stevens and Egan, 1941; Gaeth and Norris, 1965), and in normal-hearing individuals with a temporary threshold shift due to noise exposure (Webster and Schubert, 1954; Ward, Selters and Glorig, 1961). Diplacusis could produce large ipsilateral–contralateral pitch-match differences.

Octave confusions and sources of tinnitus pitch-matching variability

Octave confusions, which have been noted previously in the pitch-matching of tinnitus (Fowler, 1942; Graham and Newby, 1962), might arise from a number of sources. A narrow-band or pure-tone tinnitus might produce its own harmonics (cf. Stevens, 1935), and therefore octave confusions may occur where the pure-tone stimulus is matched to a harmonic of the tinnitus. Where the internal tinnitus spectrum is complex, with multiple peaks that cover a wide range of frequencies, an individual could place his original pitch match at any of the corresponding frequencies. Some octave confusion may simply represent the difficulty in matching the single pitch of the tone to a complex tinnitus without any clear prominent pitch. With acoustic signals, the more diffuse the spectrum the more subtle is the pitch (Fastl and Stoll, 1979).

If we assume that tinnitus resembles a wide-band noise, then data from normal listeners suggest that such a spectrum can produce pitches in the middle of the band or at the spectral edges (Ekdahl and Boring, 1934; Bekesy, 1961; Small and Daniloff, 1967). Thus, different people may match the pitch to the upper edge, centre frequency, or lower edge of their tinnitus spectra.

The fact that we failed to observe any octave confusions in the ear contralateral to the tinnitus may suggest that the octave confusion is produced by some peripheral interaction between the tinnitus and the pure tone. It is also interesting that 21 out of the 26 octave confusions resulted in a decision to change to a lower pitch than the original match (although in about a quarter of our pitch matches it was not possible to present a tone one octave above the pitch-match frequency which was audible to the subject). Further work is required to substantiate these observations.

Tentative clinical recommendations

Although tinnitus pitch matching is imprecise in some patients, in others it is reasonably repeatable. Therefore, we suggest that tinnitus pitch matching has potential for the quantification of tinnitus.

It is noteworthy that three subjects (1,4,7) in our present experiment returned 3 to 4 months later to participate in a tinnitus-loudness experiment (Tyler and Conrad-Armes, 1983). The tinnitus pitch-match frequency was determined for this later experiment using the Method of Adjustment with three replications, and the results were all within $\frac{1}{3}$ octave of the earlier measurements. This suggests that tinnitus pitch matches can be quite stable over time in some cases and that tinnitus pitch could be used to monitor changes in tinnitus characteristics, either clinically or experimentally.

We recommend measuring tinnitus pitch in the ear ipsilateral to the tinnitus, to avoid any effects of binaural diplacusis. The Method of Limits requires the greatest amount of time (typically 4 to 5 min) to obtain a pitch match since it requires both a descending and ascending approach, and also produced the most octave confusions. The Adaptive Method and the Method of Adjustment both require 1 to 2 min to obtain a single pitch match. Therefore, we recommend either the Adaptive Method or the Method of Adjustment for clinical use. The Adaptive Method does appear to produce fewer octave confusions. The more pitch matches that are obtained from a patient, the more certain one can be of the accuracy of the average pitch match. Considering the large individual difference in the ability to perform repeatable pitch matches, it would seem preferable to obtain as many as 7 to 9 pitch matches and determine variability.

When octave confusions do occur, they should be noted along with the direction of the octave confusion. Our present clinical practice is *not* to include trials where octave confusions occur in our average pitch-match calculations, but instead to repeat the pitch-match trial.

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